Sahaj Bhadja

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Summary

Software engineer with 2+ years building **autonomous systems**, **ML-based planning algorithms**, and **simulation pipelines**. Experience with trajectory planning for autonomous driving, data analysis on large-scale datasets, and deploying robust production systems in C++ and Python.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Robotics, Machine Learning, Linear Algebra, Data Structures

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Control Systems, Numerical Methods, Algorithms, Object-Oriented Programming

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **robust backend systems in Python** with fault-tolerant async processing, implementing structured state transitions and failure isolation that increased throughput by **3x** under concurrent workloads.
- Developed **data analysis pipelines** processing 5K+ monthly jobs with SQL optimization, validation logic, and performance tuning, achieving 99%+ reliability in production environments.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed **concurrent batch-processing systems** using multithreading, processing 1K+ files per batch with fault-tolerant execution and thorough analysis of system performance under varying loads.
- Wrote **production-quality code in Python** with modular architecture, participating in code reviews and collaborating across teams to rapidly iterate on complex software projects.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a **GPT-based Decision Transformer for autonomous vehicle planning** using PyTorch, implementing offline reinforcement learning algorithms that improved long-horizon trajectory planning by **18%** in simulation.
- Architected a **spatial-temporal encoder integrating HD-map geometry** (road layout, lane direction) with ego-agent trajectory context via multi-head attention, improving closed-loop planning stability by 15% in simulation evaluation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform software with complex systems integration**, implementing algorithms that improved response times by 30% and pipeline efficiency by **68%** through structured problem decomposition.
- Built **automated testing and CI/CD pipelines** using Jenkins and GitHub across Windows, Linux, and macOS, delivering robust code with 20+ unit/integration test suites in Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **scalable microservice systems** handling 2M+ records with real-time synchronization, implementing efficient data pipelines and structured solutions for complex distributed problems.
- Built **Docker-based deployment infrastructure** with CI/CD automation, collaborating with engineering teams to scale systems and ensure reliable performance across production workloads.

Skills

Languages: Python, C++, C, SQL, JavaScript/TypeScript, Bash
ML/Robotics: PyTorch, Transformers, offline RL, trajectory planning, simulation pipelines, data analysis
Infrastructure: Docker, AWS, GCP, Linux, Git, CI/CD, distributed systems
Practices: Algorithm design, systems integration, Agile/Scrum, technical documentation

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built a **semantic retrieval system** using transformer embeddings and vector similarity search, processing 50K+ documents with optimized algorithms achieving sub-second retrieval and 95% query accuracy.

CPU-Scheduling Simulator Live | *C++, JavaScript, React, Algorithms*
- Implemented **scheduling algorithms** (FCFS, SJF, SRT, Round Robin, MLQ) in C++, demonstrating algorithm design skills with performance analysis showing 25% reduction in average turnaround time.

Linguistix Code | *Python, AWS (ECS, Fargate, CodePipeline), Docker*
- Developed a **containerized Python application** with automated CI/CD pipelines, implementing validation checks and achieving production-grade reliability through systematic testing and deployment.